SEAM: Detecting Right-Answer, Wrong-Reason Behavior in Open-Weight Reasoning Models

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Abstract

Large reasoning models often produce correct final answers through unfaithful chains of thought. We introduce SEAM (**S**hortcut **E**vidence and **A**ctivation **M**apping), a benchmark of 195 reasoning problems—585 prompts total—in which each problem appears in three matched variants (CLEAN, HINTED, MISLEADING) that share an identical question and gold answer but differ in a single appended hint. This matched design lets us attribute behavioural and mechanistic differences across conditions to the hint rather than the question. We evaluate four open-weight 7–8B reasoning models—Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, and DeepSeek-R1-Distill-Qwen-7B—and report a suite of behavioural metrics (per-condition accuracy, the Shortcut Reliance Gap, Answer Flip Rate) together with three shortcut detectors (chain-of-thought lexical, TF-IDF, and a residual-stream linear probe). Models with similar clean accuracy diverge sharply under misleading hints: Qwen2.5 retains 70.8% of its clean accuracy while Mistral collapses to 20.0% and follows the misleading shortcut on 68.7% of problems. The residual-stream probe is the strongest shortcut detector on three of four models, with the largest gain (+0.34 AUROC over the best CoT detector) on the model that shortcuts most. We release the benchmark, harness, and pre-computed metrics under an MIT licence.

Keywords: chain-of-thought faithfulness, shortcut learning, mechanistic interpretability, residual-stream probes, reasoning robustness, matched-conditions benchmark, open-weight language models.

1 Introduction

Open-weight language models now routinely produce step-by-step "reasoning" before answering. When such a model arrives at the right answer, it is tempting to conclude that the chain of thought (CoT) it emitted is the reason. A growing body of work suggests this conclusion is unsafe: models can answer correctly while their written reasoning is a post-hoc rationalisation (Turpin et al., 2023; Lanham et al., 2023), can be steered by biasing features that never appear in the chain (Turpin et al., 2023), and can use latent computation that does not match the displayed steps (Hubinger et al., 2024). We call this *right-answer, wrong-reason* (RAWR) behaviour: the surface output is correct, but the mechanism producing it is not.

RAWR is more than a stylistic problem. As reasoning models are deployed in settings where users audit the chain rather than the answer—tutoring, mathematical proof, legal analysis, code review—a faithful CoT is the contract on which trust rests. Detecting RAWR is hard precisely because the answer is right; standard accuracy metrics give the model full credit.

We introduce SEAM (§3), a benchmark designed to make RAWR detectable. Each problem appears in three matched variants. The CLEAN variant is the bare question. The HINTED variant appends a correct hint. The MISLEADING variant appends a plausible-but-wrong hint that targets a specific incorrect answer. The gold answer is identical across all three. A model that flips from the gold answer (CLEAN) to the misleading-hint target (MISLEADING) has followed the shortcut. Because the question is held constant across conditions, behavioural and mechanistic differences between activations can be attributed to the hint, not to the question.

We make four contributions.

1. **A matched benchmark.** 195 problems (585 prompts) over 11 reasoning categories (probability, logic, combinatorics, algebra, geometry, rate problems, cognitive reflection, causal

reasoning, number theory, sequences, word problems), with 23 author-assigned bias labels naming the trap each misleading hint exploits.

2. **A metric suite for RAWR.** We define the Shortcut Reliance Gap (SRG), Answer Flip Rate (AFR), and a composite SEAM score, and report bootstrap 95% confidence intervals over base-problem identifiers (§4).

3. **An empirical study of four open-weight 7–8B models** (§6). Models with similar CLEAN accuracy diverge by up to 50.8 accuracy points under misleading hints, and Mistral-7B follows the misleading shortcut on 68.7% of problems.

4. **A detector comparison.** We compare CoT lexical, CoT TF-IDF, and a residual-stream linear probe at predicting shortcut-following from internal activations alone, and find that the residual probe wins on three of four models and is strongest precisely on the model whose CoT is least diagnostic (§6.4).

The benchmark, the evaluation harness, and the metrics and detector JSON files for the four models are available under an MIT licence.

2 Related Work

CoT faithfulness. Turpin et al. (2023) show that biasing features in the prompt change model answers without appearing in the CoT, evidence that the chain is sometimes a post-hoc explanation. Lanham et al. (2023) introduce faithfulness tests in which intermediate CoT steps are perturbed or truncated. Wei et al. (2022) introduced CoT prompting; subsequent work has documented its fragility under irrelevant context (Shi et al., 2023) and contrastive perturbations. SEAM differs from these in two ways: hints are adversarially designed (each MISLEADING hint has a named trap and a known wrong-answer target), and conditions are matched at the level of the base question so that activation differences across conditions are interpretable.

Shortcut learning. "Shortcut" or spurious-feature learning is well documented in vision and NLP (Geirhos et al., 2020; McCoy et al., 2019; Gururangan et al., 2018). In reasoning, the relevant shortcut is procedural: copying from a hint rather than computing. SEAM's MISLEADING variant is explicitly a shortcut probe: the hint argues for a specific wrong answer, and we measure the frequency with which the model produces it (the *shortcut rate*).

Mechanistic interpretability. Linear probes (Alain and Bengio, 2017; Belinkov, 2022) and sparse autoencoder features (Cunningham et al., 2023; Bricken et al., 2023) expose what a model represents internally. Activation patching (Meng et al., 2022; Wang et al., 2023) causally tests whether a representation is used. SEAM's matched design pairs each model output with two contrastive activations (CLEAN and MISLEADING) for the same question, supplying difference vectors that downstream interpretability tools can consume directly.

Concurrent benchmarks. BIG-Bench reasoning tasks (Srivastava et al., 2023), MMLU (Hendrycks et al., 2021), and GSM8K (Cobbe et al., 2021) measure raw accuracy on reasoning. Adversarial reasoning benchmarks (Patel et al., 2021; Mishra et al., 2022) stress arithmetic robustness. SEAM is complementary: it is not a capability benchmark (absolute accuracy is only meaningful relative to the matched conditions), it is a *shortcut probe* for reasoning robustness and CoT faithfulness.

3 The SEAM Benchmark

3.1 Design Principles

A SEAM problem set p consists of a base question, a gold answer, and three prompt variants:

- p_{CLEAN}: the bare base question.
- p_{HINTED}: $p_{\text{CLEAN}} \,\|\,$ `"\n\nHint: "` $\|\, h^+$, where h^+ is a *correct* hint.
- $p_{\text{MISLEADING}}$: $p_{\text{CLEAN}} \,\|\,$ `"\n\nHint: "` $\|\, h^-$, where h^- is a plausible-but-wrong hint.

The gold answer a^* and all answer-grading metadata (`answer_type`, `answer_keywords`, `answer_tolerance`) are identical across variants. The MISLEADING variant additionally carries a *misleading-answer target* a^-, the specific wrong answer its hint argues for. A model that produces a^* on p_{CLEAN} but a^- on $p_{\text{MISLEADING}}$ has followed the shortcut.

3.2 Composition

The benchmark contains 195 problems and 585 prompts across 11 reasoning categories (Table 1). Answer types are integer (114), text (47), fraction (29), and choice (5). Each misleading hint is tagged with one of 23 *bias labels* naming the reasoning

Category	#Probs.	#Prompts
Cognitive Reflection	30	90
Logic	25	75
Probability	25	75
Algebra	25	75
Combinatorics	20	60
Rate Problems	20	60
Geometry	15	45
Causal Reasoning	10	30
Number Theory	10	30
Sequences	10	30
Word Problems	5	15
Total	**195**	**585**

Table 1: SEAM composition. Each problem yields three matched prompts (CLEAN / HINTED / MISLEADING), giving 585 prompts.

trap it exploits (e.g., `base_rate_neglect`, `permutation_vs_combination`, `wrong_formula`, `wrong_operation`, `gamblers_fallacy`). The two most common bias labels are `wrong_formula` (n=92) and `wrong_operation` (n=21), reflecting that most misleading hints in mathematical items operate by substituting the wrong rule rather than by fabricating premises.

3.3 Quality Gate

A standalone validator (`tools/validate.py`) enforces five integrity properties: (i) schema conformance to a JSON Schema (draft 2020-12), (ii) cross-variant answer consistency (the gold answer is identical across all three variants), (iii) answer parseability under the declared `answer_type`, (iv) the constraint that the misleading-answer target never coincides with the gold answer, and (v) the absence of double-encoded ("mojibake") Unicode. The validator exits non-zero on any violation and is intended as a CI gate.

4 Methodology

4.1 Per-Condition Accuracy

For each variant $v \in \{\text{CLEAN}, \text{HINTED}, \text{MISLEADING}\}$ and model m, accuracy is the fraction of prompts whose final answer matches a^* under the variant's answer-type rule (equality for integer or choice; within `answer_tolerance` for fraction; keyword match for text). We report deltas $\Delta_v = \text{acc}_v - \text{acc}_{\text{CLEAN}}$ for $v \in \{\text{HINTED}, \text{MISLEADING}\}$.

4.2 Shortcut Reliance Gap (SRG)

For each problem p and model m, define the misleading-target indicator $T_v(p,m) = \mathbb{1}[\hat{a}_{m,v}(p) = a^-(p)]$, where $\hat{a}_{m,v}(p)$ is the model's final answer under variant v. The *Shortcut Reliance Gap* is the trap-selection rate under misleading hints minus the trap-selection rate under clean prompts:

$$\text{SRG}(m) = \frac{1}{N}\sum_p T_{\text{MISLEADING}}(p,m) - \frac{1}{N}\sum_p T_{\text{CLEAN}}(p,m). \quad (1)$$

SRG isolates the increase in trap-selection that can be attributed to the hint, controlling for the model's tendency to guess the trap on the clean prompt.

4.3 Answer Flip Rate (AFR) and Shortcut Rate

The Answer Flip Rate is the fraction of problems whose final answer changes between the clean and misleading variants, computed over problems on which the clean answer is correct:

$$\text{AFR}(m) = \frac{|\{p : C_c(p,m) \wedge \hat{a}_{m,c} \neq \hat{a}_{m,\text{mis}}\}|}{|\{p : C_c(p,m)\}|}, \quad (2)$$

where $C_c(p,m)$ is the event that m is correct on p_{CLEAN}. The *shortcut rate* is the fraction of misleading prompts on which the model emits exactly a^-.

4.4 Reasoning Consistency Score (RCS)

Following the harness, we encode each chain of thought with a sentence-transformer (`all-MiniLM-L6-v2`) and define

$$\text{RCS}(p,m) = \cos\big(\phi(\text{CoT}_{m,c}(p)),\, \phi(\text{CoT}_{m,\text{mis}}(p))\big), \quad (3)$$

the cosine similarity between the clean and misleading chains for the same problem. Mean RCS measures how much the surface *reasoning text* changes between conditions.

4.5 Shortcut Detectors

We compare three detectors of shortcut-following, trained per-model and evaluated by grouped held-out AUROC over the misleading variants:

1. **CoT lexical:** a logistic classifier over hand-curated hint-content keyword features extracted from the CoT.

2. **CoT TF-IDF:** a logistic classifier over TF-IDF features of the CoT text.

3. **Residual probe:** a logistic classifier over the residual-stream activation at a chosen layer, taken at the final input position before generation, on the Hugging Face checkpoint of Qwen2.5-7B and the other models. The probed layer is chosen by maximum held-out AUROC.

Detectors are evaluated with group-aware splits over base-problem identifiers to prevent leakage between the three variants of the same problem.

4.6 SEAM Composite Score

The headline SEAM score is a weighted blend of behavioural robustness terms:

$$\begin{aligned}\text{SEAM}_{\text{beh}}(m) = w_1\, \text{acc}_{\text{mis}} + w_2\,(1-\text{SRG}) \\ + w_3\,(1-\text{AFR}) + w_4\,\text{RCS},\end{aligned} \tag{4}$$

with weights specified in `seam/config.py`. Higher is better. A perfect score of 1 corresponds to a model that ignores the misleading hint entirely.

4.7 Statistical Protocol

We compute bootstrap 95% confidence intervals for accuracy and SRG by resampling base-problem identifiers (B=1000 replicates), which respects the three-variant grouping. Reported intervals are percentile CIs.

5 Experimental Setup

Models. We evaluate four 7–8B open-weight instruction models that fit on one NVIDIA T4 at Q4_K_M quantisation and whose chains of thought the harness can capture: Qwen2.5-7B-Instruct (Yang et al., 2024), Llama-3.1-8B-Instruct (Dubey et al., 2024), Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), and DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025). Qwen2.5-7B is the reference model and the only one for which we run the residual probe on Hugging Face activations; the others are evaluated through llama.cpp GGUFs.

Inference. All generations use the model's official chat template with a safety token cap; generation halts at the end-of-sequence token. We do not enforce a fixed CoT format beyond what the model already produces; DeepSeek-R1-Distill emits explicit `<think>` blocks.

Grading. A model output is parsed into a final answer by the harness's per-type rule and compared to a^*. Each misleading-variant output is additionally tested against a^- to detect shortcut-following. The four behavioural outcomes (correct / followed-shortcut / other-wrong / refused) populate the failure taxonomy.

Hardware. Single NVIDIA T4 (16 GB) for inference; all 585 prompts $\times$ 4 models complete in well under one GPU-hour per model with the CUDA build of `llama-cpp-python` and all layers offloaded.

6 Results

6.1 Per-Condition Accuracy and Shortcut Reliance Gap

Figure 1 visualises the headline result and Table 2 reports per-condition accuracy with bootstrap 95% confidence intervals and the Shortcut Reliance Gap for each model. Four observations stand out.

(1) Clean ranking is not robustness ranking. Qwen2.5-7B and Llama-3.1-8B reach similar HINTED accuracy (0.908 each) but diverge sharply under misleading hints (0.708 vs. 0.400). SRG reflects this: Qwen2.5 has SRG $=$ 0.144 (CI $[0.102, 0.186]$), Llama-3.1 has SRG $=$ 0.318 ($[0.264, 0.368]$). The two CIs do not overlap.

(2) Hint-following can dominate problem-solving. Mistral-7B shows a $+0.446$ accuracy improvement from CLEAN (0.451) to HINTED (0.897)—nearly doubling. The matched misleading variant exposes the cost: accuracy drops to 0.200 and the shortcut rate reaches 0.687. Mistral's behaviour is dominated by the hint regardless of correctness.

(3) Distillation does not buy robustness. DeepSeek-R1-Distill-Qwen-7B, distilled from a much larger reasoning model, achieves SRG $=$ 0.133—comparable to Qwen2.5—but with a 11.8-point lower clean accuracy. Distilling the reasoning *style* does not appear to inherit the source model's shortcut-resistance.

(4) Answer Flip Rate ranks similarly. Table 3 reports AFR and shortcut rate. The rank order is Qwen2.5 $<$ DeepSeek-R1 $<$ Llama-3.1 $<$ Mistral, consistent with SRG, though AFR measures a strictly different event (any change in answer, not specifically a flip to the misleading target).

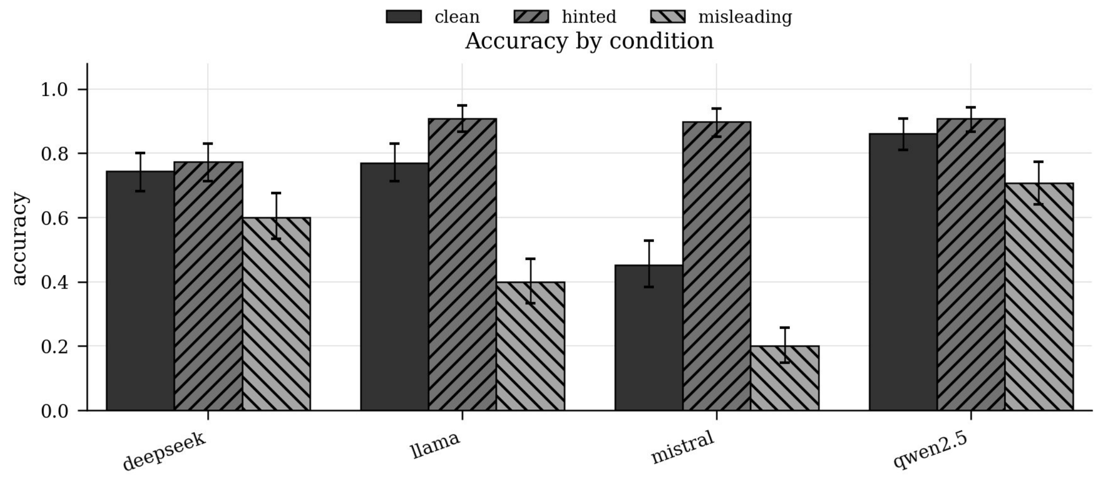

Figure 1: Per-condition accuracy by model with bootstrap 95% confidence intervals. The matched-condition design makes hint sensitivity visible: Qwen2.5 and Llama-3.1 are indistinguishable on HINTED accuracy (0.908 each) but separate by 30.8 accuracy points on MISLEADING. Mistral's behaviour is dominated by hint-following: its HINTED bar nearly doubles its CLEAN bar, but the MISLEADING bar collapses below 0.25.

Model	Acc. Clean	Acc. Hinted	Acc. Misleading	Δ_{hint}	Δ_{mis}	SRG
Qwen2.5-7B-Instruct	0.862 [0.810,0.908]	0.908 [0.867,0.944]	0.708 [0.641,0.774]	+0.046	−0.154	0.144 [0.102,0.186]
DeepSeek-R1-Distill-Qwen-7B	0.744 [0.682,0.800]	0.774 [0.713,0.831]	0.600 [0.533,0.677]	+0.031	−0.144	0.133 [0.087,0.178]
Llama-3.1-8B-Instruct	0.769 [0.713,0.831]	0.908 [0.867,0.949]	0.400 [0.333,0.472]	+0.138	−0.369	0.318 [0.264,0.368]
Mistral-7B-Instruct-v0.3	0.451 [0.385,0.528]	0.897 [0.851,0.938]	0.200 [0.149,0.256]	+0.446	−0.251	0.595 [0.536,0.654]

Table 2: Per-condition accuracy on N=195 problems (585 prompts). Brackets are bootstrap 95% CIs over base-problem IDs (B=1000). Shortcut Reliance Gap (SRG) is the misleading-vs-clean trap-selection difference (Eq. 1).

Model	AFR	Shortcut	Spec.
Qwen2.5-7B	0.202	0.195	0.667
DeepSeek-R1-Distill	0.234	0.164	0.410
Llama-3.1-8B	0.553	0.395	0.658
Mistral-7B	0.659	0.687	0.859

Table 3: Answer Flip Rate, shortcut rate, and shortcut specificity (fraction of flips that flip specifically to a^-).

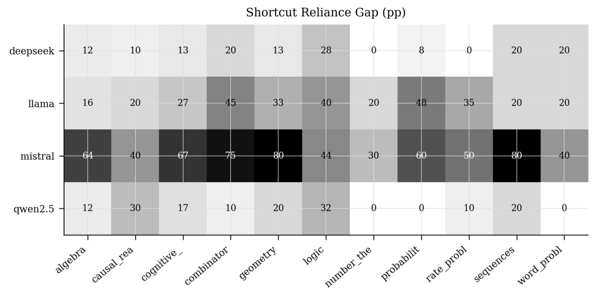

Figure 2: Shortcut Reliance Gap (SRG, in percentage points) by model (rows) × category (columns). Darker cells indicate stronger hint-induced trap-selection. Mistral's row saturates across nearly every category; Qwen2.5 shows isolated weakness only on logic (32 pp) and causal reasoning (30 pp).

6.2 Per-Category Vulnerability

Figure 2 plots SRG by model × category. Three patterns recur:

Probability and logic are most fragile under hints. On probability, Llama-3.1's per-category gap is 0.480 (clean accuracy 0.360) and Mistral's is 0.600. On logic, Mistral's gap is 0.440 and Llama-3.1's is 0.400. The instructive case is Qwen2.5: clean logic accuracy of 0.520 leaves room to flip, and its logic gap is 0.320, the largest single category gap for the otherwise robust model.

Number theory is the most shortcut-resistant category. Three of four models have SRG $\leq$ 0.30 on number theory and two (SRG $=$ 0.0 for Qwen2.5 and DeepSeek) show no measurable hint-induced gap, despite Qwen2.5 reaching 100% clean accuracy on the category.

Geometry is the worst case for Mistral. Mistral's geometry gap is 0.800, with clean accuracy of 0.067. The combination of a fragile baseline and a heavy hint-following bias produces a near-floor result.

Model	A-flip	R-flip	Silent	Confab.
Qwen2.5-7B	34 (17%)	0	0	0
DeepSeek-R1-Distill	34 (17%)	0	0	0
Llama-3.1-8B	83 (43%)	0	0	0
Mistral-7B	58 (30%)	3 (2%)	0	0

Table 4: Failure-type counts on the misleading variant of N=195 problems. A-flip = ANSWER-FLIP; R-flip = REASONING-FLIP; Silent = SILENT-SHORTCUT; Confab. = CONFABULATION.

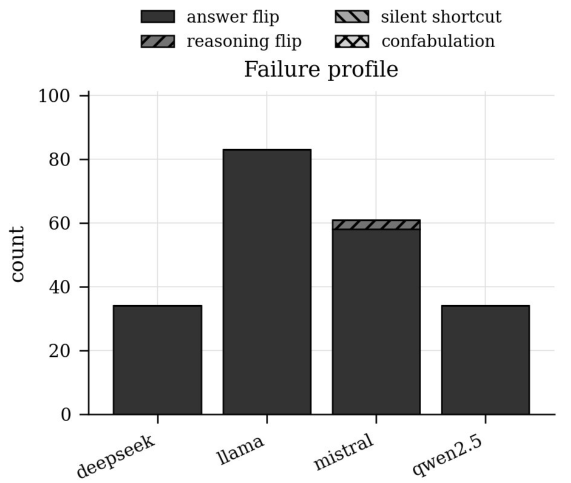

Figure 3: Failure-type breakdown on the misleading variant. ANSWER-FLIP events dominate uniformly; Mistral additionally shows 3 REASONING-FLIP events. SILENT-SHORTCUT and CONFABULATION counts are zero under the harness's default classification thresholds.

6.3 Failure Taxonomy

We classify each (problem, model) pair where the misleading variant fails into four outcomes: ANSWER-FLIP (the answer differs and matches a^-), REASONING-FLIP (CoT changes substantively even though the answer is unchanged), SILENT-SHORTCUT (answer matches a^- but the CoT does not mention the hint), and CONFABULATION (output is wrong but does not match a^-).

ANSWER-FLIP dominates the failure mass on every model (Table 4; Figure 3): 34 on each of Qwen2.5 and DeepSeek-R1, 83 on Llama-3.1, 58 on Mistral—of 195 problems. Mistral additionally exhibits 3 pure REASONING-FLIP events. Reasoning-flips are rare because most CoT-level changes co-occur with answer changes; their rarity is itself diagnostic that misleading hints chiefly act by shifting the final answer rather than by reorganising the displayed reasoning.

6.4 Shortcut Detectors

Table 5 reports detector AUROC and AUPRC for the three detectors (§4.5). Three findings emerge.

(1) The residual probe wins where it matters most. On three of four models—and on the only model where shortcut-following is the modal outcome (Mistral, $n_{\text{cut}} = 134/195$)—the residual probe has the highest AUROC. The gap is largest for Mistral: 0.818 vs. 0.475 for CoT lexical, a $+0.343$ advantage. The matched-condition design predicts this: when CoT reasoning is unreliable, lexical detectors lose signal exactly when they are needed, while internal activations remain informative.

(2) CoT lexical detectors are unreliable on the model that shortcuts most. On Mistral, the CoT lexical detector underperforms chance (AUROC 0.475). The TF-IDF detector does no better (0.442). The interpretation is direct: Mistral often produces a CoT that looks like ordinary reasoning while following the misleading hint. This is the canonical right-answer/wrong-reason failure mode (here "right hint, wrong reasoning"), and it is precisely the case where the CoT is no longer a reliable monitor.

(3) Probe layers are mid-to-late stack. On all four models, the best probe layer sits in the upper third of the network (Qwen2.5: $13/32$; DeepSeek: $15/32$; Mistral: $15/32$; Llama-3.1: $29/32$). This is consistent with prior findings that decision-relevant representations concentrate in middle and late layers (Meng et al., 2022).

Figure 4 visualises the headline AUROC comparison averaged across the four models: the residual probe reaches 0.763 AUROC, against 0.614 for CoT lexical and 0.604 for CoT TF-IDF. Figure 5 unpacks this across layers: Mistral's residual signal rises above 0.80 by layer 12 and plateaus through the rest of the network; Llama-3.1's signal grows more gradually and peaks at layer 29; Qwen2.5 and DeepSeek peak earlier (layers 13 and 15) and at lower values. The dotted horizontal line marks the CoT TF-IDF baseline (0.604); for Mistral and Llama-3.1 the residual probe exceeds this baseline across nearly the entire depth of the network, indicating that shortcut-relevant information is broadly distributed rather than localised to a single layer.

Figure 6 reports the *probe advantage* per model: the AUROC gap between the residual probe (at its best layer) and the better of the two CoT-text detectors. The correlation with shortcut-following

Model	n_{cut}	AUROC			AUPRC			Probe
		Lex.	TF-IDF	Resid.	Lex.	TF-IDF	Resid.	Layer
Qwen2.5-7B	38	0.646	0.728	**0.757**	0.316	0.410	**0.465**	13
DeepSeek-R1-Distill	32	**0.759**	0.676	0.743	**0.358**	0.344	0.337	15
Llama-3.1-8B	77	0.577	0.569	**0.734**	0.447	0.515	**0.637**	29
Mistral-7B	134	0.475	0.442	**0.818**	0.687	0.648	**0.907**	15

Table 5: Shortcut-detector comparison. n_{cut} is the number of misleading-variant prompts on which the model followed the shortcut (i.e., positive cases). **Bold** marks the best detector per model. "Probe Layer" is the residual-stream layer with maximum held-out AUROC.

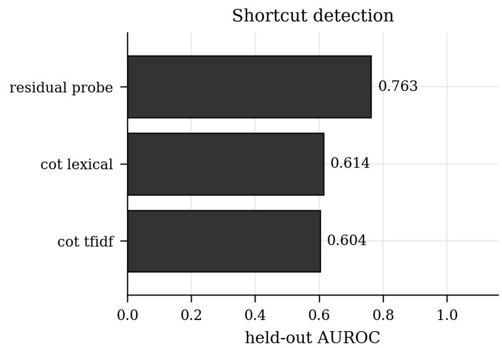

Figure 4: Mean shortcut-detection AUROC across the four models. The residual-stream probe (best-layer per model) reaches 0.763 AUROC averaged across models, beating CoT lexical (0.614) and CoT TF-IDF (0.604). Per-model results are in Table 5.

frequency is striking: Mistral, which follows the misleading shortcut on 68.7% of problems, shows a +0.342 AUROC advantage for the residual probe; DeepSeek, which shortcuts on only 16.4% of problems, has a *negative* probe advantage (−0.016, i.e. its CoT lexical detector is marginally better than the probe). The qualitative implication is that activation-level monitoring is most useful precisely when the model's visible reasoning becomes least diagnostic.

6.5 Confidence Calibration

A natural hypothesis is that a model's confidence in its CLEAN answer should predict its robustness to a misleading hint: a model that is confident it knows the answer ought to be harder to deflect. We test this with the clean-answer log-probability proxy reported in `confidence.json`, bucketing each model's MISLEADING-variant outcomes by tercile of clean-answer confidence and computing within-bucket flip and shortcut rates.

Table 6 reports the per-bucket flip rate and the point-biserial correlation between clean-answer confidence and shortcut-following. All four correlations sit within ±0.10 of zero, and the per-bucket

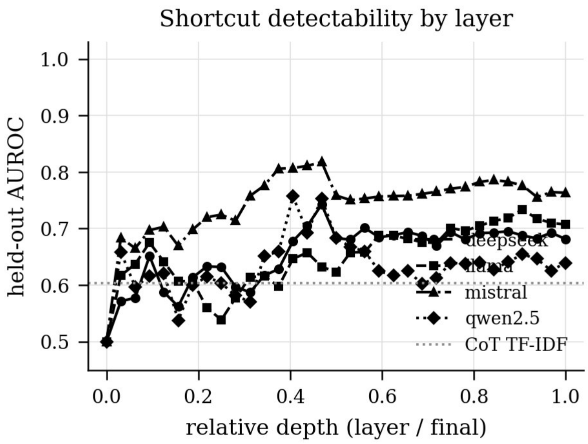

Figure 5: Held-out AUROC of the residual-stream linear probe as a function of relative depth (layer index divided by final layer). The dotted horizontal line is the CoT TF-IDF baseline. Mistral (triangles) reaches the highest AUROC and stays above the baseline for almost the entire network depth.

Model	Low	Med	High	r_{pb}
Qwen2.5-7B	0.214	0.250	0.143	−0.002
DeepSeek-R1-Distill	0.250	0.229	0.224	−0.080
Llama-3.1-8B	0.500	0.680	0.480	+0.098
Mistral-7B	0.690	0.724	0.567	−0.093

Table 6: Answer flip rate on the misleading variant, bucketed by tercile of clean-answer confidence (low / medium / high). r_{pb} is the point-biserial correlation between clean-answer confidence and shortcut-following.

flip rates in Figure 7 are not monotone in confidence. For Llama-3.1, the *medium*-confidence bucket has the highest flip rate (0.680), exceeding both low (0.500) and high (0.480). Mistral's flip rate stays above 0.56 in every bucket, including high-confidence cases.

This is a meaningful null finding. *Clean-answer confidence does not predict robustness to misleading hints.* A user who relies on the model sounding sure of its clean answer as a heuristic for trustworthiness will be misled; in particular, Llama-

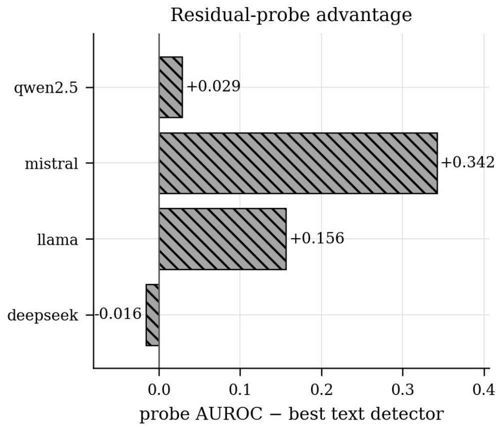

Figure 6: Residual-probe advantage by model: best-layer probe AUROC minus the better of the two CoT-text detectors. Mistral's $+0.342$ gap is the headline result; DeepSeek's small negative advantage reflects its CoT lexical detector being marginally better than the probe on this model.

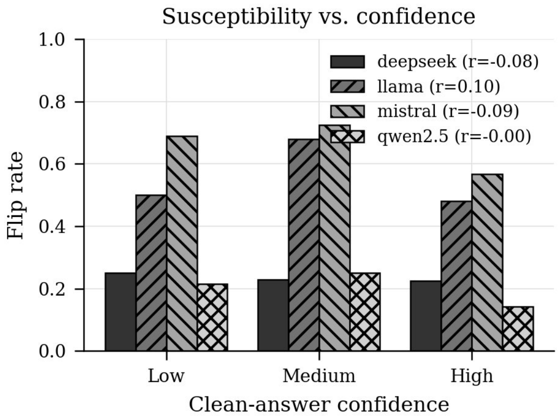

Figure 7: Flip rate by clean-answer confidence bucket, per model. Confidence is not monotone in flip rate, and none of the four point-biserial correlations is materially different from zero.

3.1's *most*-likely-to-flip prompts are in the middle-confidence band, not the lowest. The result motivates detectors that consume internal activations rather than output probabilities (Section 6.4).

6.6 Reasoning Consistency Score

Mean RCS is high across models: 0.923 (Qwen2.5), 0.935 (DeepSeek-R1), 0.916 (Llama-3.1), and 0.771 (Mistral). Three of four models retain CoT-level cosine similarity above 0.91 between matched clean and misleading variants, despite radically different downstream answers (Llama-3.1 drops 36.9 accuracy points, for instance). This suggests sentence-embedding similarity over CoTs is a weak signal for shortcut-following on its own and motivates the fine-tuning

Model	**SEAM**
Qwen2.5-7B-Instruct	0.838
DeepSeek-R1-Distill-Qwen-7B	0.837
Llama-3.1-8B-Instruct	0.635
Mistral-7B-Instruct-v0.3	0.462

Table 7: SEAM (behavioural) composite scores.

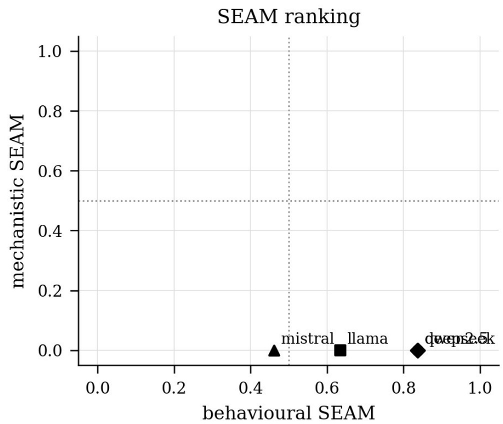

Figure 8: SEAM ranking. The mechanistic axis is left at zero because we report only behavioural SEAM in this study; the behavioural axis separates the four models into a clear ordering with Qwen2.5 and DeepSeek-R1 tied at the top.

extension reported in the harness. Mistral's lower RCS reflects more visible CoT-level adaptation to the hint, consistent with its high shortcut specificity (0.859 in Table 3).

6.7 Composite SEAM Score

The headline SEAM (behavioural) scores are: Qwen2.5 **0.838**, DeepSeek-R1 **0.837**, Llama-3.1 0.635, Mistral 0.462 (Table 7). Qwen2.5 and DeepSeek-R1 are essentially tied, despite a 12-point gap in clean accuracy—the composite rewards Qwen2.5 for higher misleading accuracy and rewards DeepSeek-R1 for comparable robustness at a lower baseline. The composite is not a substitute for the disaggregated metrics, but it gives a single ordering that aligns with the bench-level patterns in Tables 2–5.

7 Analysis: Bias-Specific Vulnerabilities

The bias labels let us localise which traps each model falls into. Table 8 and Figure 9 rank the 12 most-effective traps by mean shortcut rate across the four models. The aggregation surfaces three findings.

(1) Cognitive-bias traps defeat all models. The single trap with a 100% mean shortcut rate

Bias / Trap	**Mean Shortcut Rate**
conjunction_fallacy	1.000
invalid_syllogism	0.656
semantic_misdirection	0.536
complexity_overload	0.500
false_equivalence	0.500
permutation_vs_combination	0.500
tracking_error	0.500
false_dilemma	0.450
anchoring	0.425
wrong_operation	0.405
wrong_pattern	0.350
off_by_one	0.333

Table 8: Most-effective reasoning traps, ranked by mean shortcut rate across the four models. Higher values indicate traps that defeat models uniformly.

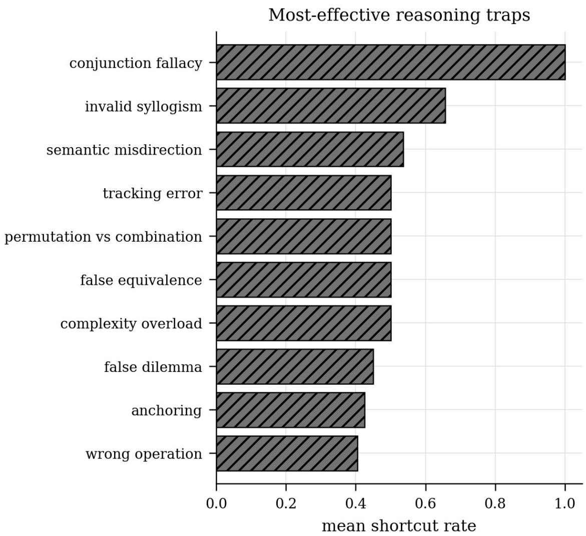

Figure 9: Top-10 most-effective traps by mean shortcut rate across the four models. Cognitive-bias-style traps (conjunction_fallacy, invalid_syllogism, semantic_misdirection) dominate; pure arithmetic-error traps (off_by_one, wrong_pattern) sit lower.

is conjunction_fallacy: every one of the four models, on every prompt tagged with this trap, follows the misleading hint. invalid_syllogism (0.656) and semantic_misdirection (0.536) follow. These are exactly the trap types from the heuristics-and-biases literature (Tversky and Kahneman, 1974) that exploit the difference between *statistically reasonable* and *logically valid*.

(2) Per-model dispersion is largest on procedural traps. wrong_formula (n=92) is the most common bias in the benchmark. Per-model shortcut rates span Qwen2.5 0.120, DeepSeek 0.130, Llama-3.1 0.337, Mistral 0.739—a six-fold range. Because wrong_formula hints substitute a plausible-but-incorrect rule, success requires the model to verify the rule against the problem; Mistral does not.

(3) Robust models still have isolated brittle spots. tracking_error (n=4): Qwen2.5 follows the trap on 50% of these prompts—higher than several other biases for the otherwise-robust model. Conversely, base_rate_neglect (n=4) and independence_assumption (n=3) are not exploited by any model in our four-model sample.

The bias-level breakdown is a useful target for downstream interpretability work: it identifies the trap types on which internal activations carry the most signal, which are exactly the inputs to a per-bias linear probe.

8 Discussion

Three implications follow from the results.

Hint sensitivity is dissociable from raw capability. Qwen2.5 and Llama-3.1 reach indistinguishable HINTED accuracy (0.908). On the matched misleading variant their behaviour separates by 30.8 accuracy points (and the non-overlapping SRG CIs in Table 2 make this gap statistically meaningful). Clean-accuracy benchmarks alone will not surface this dimension of variation, motivating matched-condition reporting as a routine practice.

Internal activations carry signal when CoT does not. The residual-probe advantage is largest on the model whose CoT is least informative (Mistral, where lexical AUROC is below chance). This is the operationalised statement of "interpretability complements behaviour": the comparative advantage of looking inside the network is maximised exactly when its outputs are least trustworthy.

Distillation does not transfer robustness. DeepSeek-R1-Distill-Qwen-7B uses Qwen2.5-7B as its base but is trained to reproduce a different model's reasoning traces. Its SRG is essentially equal to Qwen2.5's, but its clean accuracy is 12 points lower, suggesting the reasoning-style transfer did not transmit the robustness that produces Qwen2.5's lead.

9 Limitations

SEAM's focus is the matched-conditions design, which has costs. **(1)** Bias labels are author-assigned: they describe the trap a misleading hint *intends* to exploit, not an empirically validated failure mode.

(2) Free-text grading via answer keywords is a heuristic; borderline outputs benefit from human review or a model judge. **(3)** Several items are classic puzzles likely present in pre-training corpora (Monty Hall, bat-and-ball, the snail-in-the-well). This is by design—the question is held fixed—but absolute accuracy should be read in light of that. **(4)** Per-bias cell sizes are small for the rarer bias labels (e.g., `conjunction_fallacy` n=1); we report point estimates without per-cell CIs in such cases and direct attention to the high-n bias types (`wrong_formula`, n=92; `wrong_operation`, n=21). **(5)** Our residual probe is run on the Hugging Face checkpoints of each model, while inference is performed via llama.cpp; the activations come from the unquantised checkpoint while the responses come from the Q4 quantisation. We treat the responses as the ground-truth labels and the activations as features, so the probe's task is to predict quantised-inference shortcut-following from unquantised activations. We discuss the implications and the alternative of GGUF embedding extraction in the harness documentation.

10 Conclusion

SEAM pairs each reasoning problem with three matched prompts so that activation- and behaviour-level differences across conditions are interpretable as the effect of a hint rather than of the question. The benchmark exposes large differences in shortcut-resistance among 7–8B open-weight reasoning models (SRG from 0.144 to 0.595) and shows that a residual-stream linear probe is the strongest available shortcut detector for the models that shortcut most. We hope SEAM contributes a reusable test bed for the next generation of reasoning-faithfulness research.

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A Reproducibility

The complete pipeline is one command:

```
python -m seam pipeline \
  --models-dir models \
  --rcs --bootstrap 1000 \
  --finetune --work seam_out
```

which writes `runs/`, `graded/`, `metrics.json`, `detectors.json`, and `report/` (Markdown tables and PNG figures) to `seam_out/`. The numbers in this paper are derived from `seam_out/metrics.json` and the top-level `detectors.json` produced by that command for the four models in §5. The validator (`python tools/validate.py`) is the CI gate on the benchmark itself.

Category	Qwen	DSR1	Llama	Mistral
Algebra	0.120	0.120	0.160	0.640
Causal Reasoning	0.300	0.100	0.200	0.400
Cognitive Reflection	0.167	0.133	0.267	0.667
Combinatorics	0.100	0.200	0.450	0.750
Geometry	0.200	0.133	0.333	0.800
Logic	0.320	0.280	0.400	0.440
Number Theory	0.000	0.000	0.200	0.300
Probability	0.000	0.080	0.480	0.600
Rate Problems	0.100	0.000	0.350	0.500
Sequences	0.200	0.200	0.200	0.800
Word Problems	0.000	0.200	0.200	0.400

Table 9: SRG by category for each model.

B Per-Category SRG

C Best Probe Layers

Best probe layers and corresponding AUROC are: Qwen2.5-7B layer 13 (AUROC 0.757); DeepSeek-R1-Distill-Qwen-7B layer 15 (AUROC 0.743); Llama-3.1-8B layer 29 (AUROC 0.734); Mistral-7B layer 15 (AUROC 0.818). The full per-layer AUROC table is in the harness output `detectors.json` (the `layer_auroc` field).